Regional Income Growth and House Price Dynamics in the Netherlands

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Part 1 - Identify a Social Problem

1.1 Describe the Social Problem

The Dutch housing market has been under pressure for several years now. Between 2013 and 2023 the average price of houses has more than doubled, while the income growth in the same period considerably lagged behind (CBS, 2022). The increasing gap between housing costs and purchasing power is not only an economic phenomenon, but also a social problem; it determines who does have access to a stable living situation and who doesn't. According to economists at De Nederlandsche Bank, house prices have risen considerably faster than borrowing capacity, which has substantially worsened the affordability of a home since 2013 (De Meijier & Van Dijk, 2024).

It's striking that this development differs significantly across regions in the Netherlands. In the Randstad, prices increased much more than in the peripheral regions like Groningen or Zeeland, while the income development per region differs significantly too. Factors such as access to big employers, education level and local economic structure play an important role in this shift. And it is because of this regional variety that it's interesting to investigate the relation between income and house prices.

This project investigates the relationship between regional income growth and house price dynamics in Dutch provinces, based on CBS-data and the programming tool R. The goal is to discover in which regions the housing market detached most from the local income development.

Part 2 - Data Sourcing

2.1 Load in the data

For the income data we used nine separate CBS files, one for each year from 2015 to 2023. After cleaning each file individually (removing empty helper columns and adding a `jaar` year column), the files were combined row-wise into a single dataset with `rbind()`, ending in `datacombined15tot23 <- rbind(Data15161718, data2019, data2020, data2021, data2022, data2023)`. This income dataset was then merged with the house-price data (per province) and the population-density data, so that every observation is a province in a given year.

The house-price data came from a single CBS file with the average sale price of existing houses per province. We loaded it with `read.csv()` and stored it as `woningprijsprov`, which already has one row per province and a column for each year, so no files had to be combined here.

The population-density data came from the CBS file `regionale_kerncijfers.csv`. Because this file uses a semicolon as separator and has several title rows on top instead of a normal header, we read it with `read.csv(..., sep = ";", header = FALSE)` and saved the result as `Bev_dichtheid`. The actual cleaning of these last two datasets is done in section 3.1.

2.2 Provide a short summary of the dataset(s)

For this study three datasets from the Centraal Bureau voor de Statistiek (CBS) were used. The first dataset, Bestaande koopwoningen; verkoopprijzen prijsindex 2020=100, contains data on the variables 'gemiddelde verkoopprijs van bestaande koopwoningen en bijbehorende prijsindex op provinciaal niveau'. The second dataset, inkomen van huishoudens; huishoudenskenmerken, regio, contains information on the 'gemiddelde inkomen van huishoudens, op provinciaal niveau'. Lastly, the third dataset, bevolkingsdichtheid, contains information on the population density in the Netherlands.

All datasets are available on CBS StatLine. The dataset on the prices of houses spans from 2015 until 2025, the dataset on income only covers 2015 until 2023. The dataset on population density in the Netherlands spans from 1995 until 2025. For the analysis in this research the overlapping years of 2015-2023(or a part thereof) are studied.

Metadata: Density: Compiled from over 50 CBS surveys, with annual figures from 1995 onwards. Most figures are final, though some recent years contain provisional or preliminary data for specific topics such as social security and housing.

Average House Price: Based on notarial transaction records registered by the Kadaster, combined with WOZ valuations of all homes in the Netherlands. Figures are final immediately upon publication. Revisions only occur in exceptional cases. Note that the average sale price is not the same as the price index and should not be used as an indicator of price development.

Household Income: Based on tax and administrative records for all private households with known income. Figures for 2011–2022 are final; 2023 figures are preliminary. This table has been discontinued and replaced by an updated version.

2.3 Describe the type of variables included

The datasets mostly have social economic variables . The house price dataset measures the average sale price of existing houses and the price index of these sales. This is an indicator of the accessibility of the housing market in the Netherlands. The dataset on the income of households measures the average income of households per province, as a benchmark for wealth in that region. The dataset on population density, again, is based on administrative data. It measures the amount of inhabitants per square kilometer in every province.

The data originates from administrative sources, not interviews or questionnaires. The average house prices are based on the 'Basisregistratie Kadaster', in which every official sale of a house is recorded by law. The data on the income of households are deduced from tax returns and benefit registrations, processed by the CBS into regional stats. The data on population density is computed by dividing the amount of inhabitants officially listed by the amount of square kilometers in the region.

Part 3 - Quantifying

3.1 Data cleaning

Income data: Column names were replaced with clearer labels, footnote rows were removed, and the cleaned data was saved.

Housing prices: Values were reformatted with thousand separators and the dataset was saved.

Population density: This required extra steps because the original CBS file had multiple header rows before the actual data. Province names were extracted manually, the data was reshaped from wide to long format, filtered to exclude the national total. The end result is a table with one row per province and years as columns.

3.2 Generate necessary variables

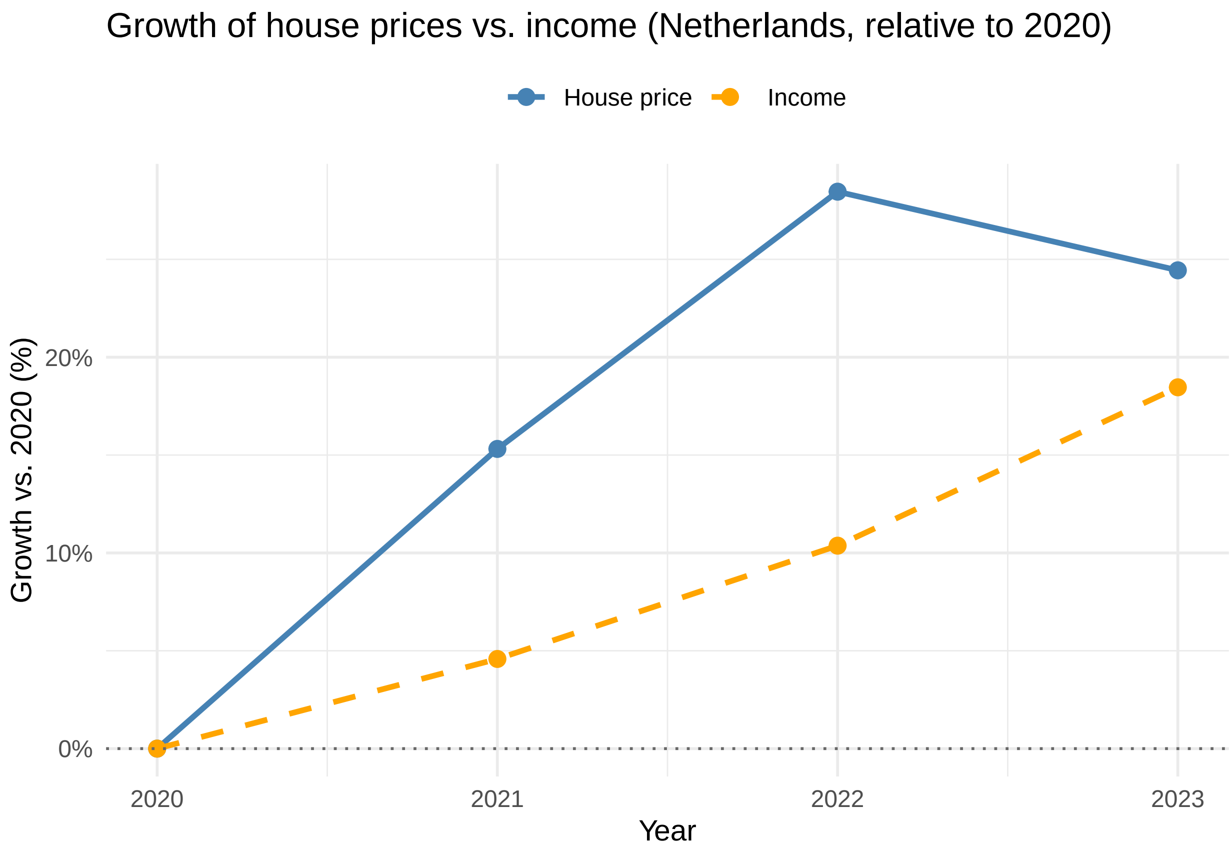
Variable 1 – average disposable income per province Our first generated variable is the average disposable household income per province, averaged over the years 2015–2023. We created it by aggregating the cleaned income dataset, taking the mean disposable income for each province. The result gives one income figure per province and serves as a stable benchmark of regional purchasing power, which we later compare against house prices.

Variable 2 – Randstad versus outside the Randstad For the second variable we grouped the provinces into two regions, the Randstad (including Flevoland) and the rest of the country, and computed the average income and average house price for each region per year. This required reshaping the house-price dataset from wide to long format and merging it with the income data, so that income and house prices could be compared on the same regional and yearly basis. This variable lets us contrast the economically dense Randstad with the other provinces.

Variable 3 – population density and house price per province The third variable that we created combines population density with the average house price for each province per year (2020–2023). We reshaped both the density data and the house-price data to long format and merged them on province and year. This variable lets us group the provinces into Randstad versus outside the Randstad and compare their house prices (used in section 3.5). We also created a fourth variable, the price-to-income ratio, which you can see later in the assignment (section 3.6).

3.3 Visualize temporal variation

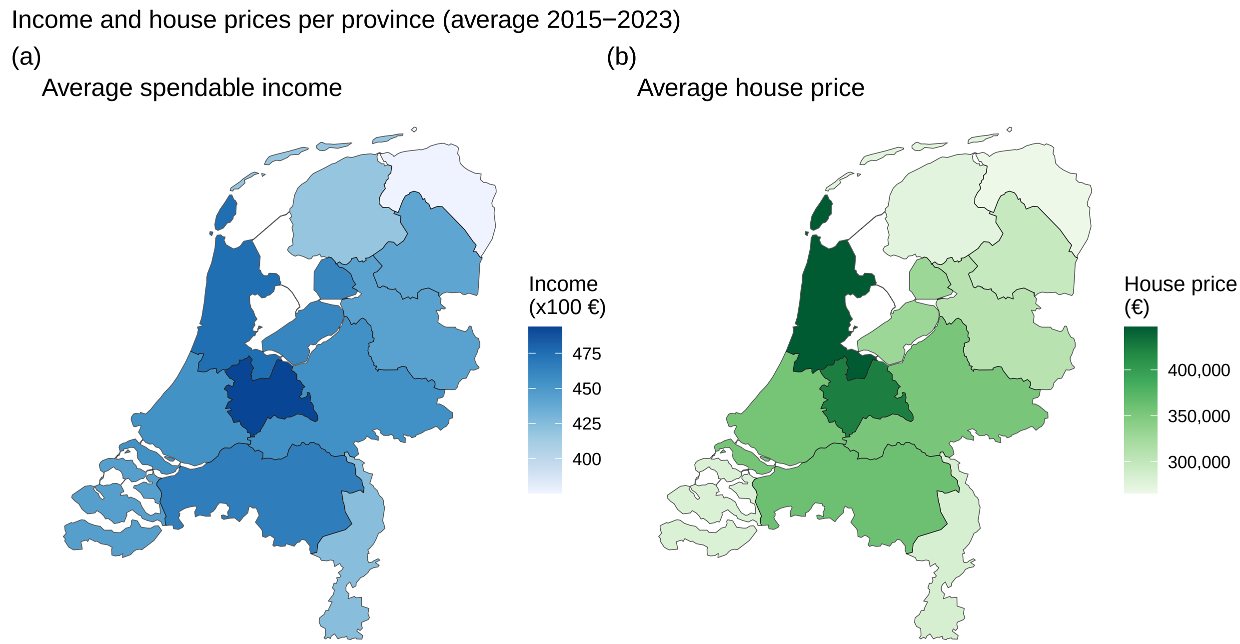
To visualize the temporal variation we averaged income and the house price across the two regions, so we get a single national figure per year, and then expressed both as growth relative to 2020. This gives us one line for house prices and one for income, which makes the widening gap between them visible in a single graph.



This graph shows the national-average growth of house prices and disposable income relative to 2020. House prices rose much faster than income: prices climbed by roughly 15% in 2021 and another 11% in 2022, while income grew by only about 4% to 5% per year. The widening gap between the two lines (the growing distance between housing costs and what people earn) is exactly the affordability problem central to this assignment.

3.4 Visualize spatial variation

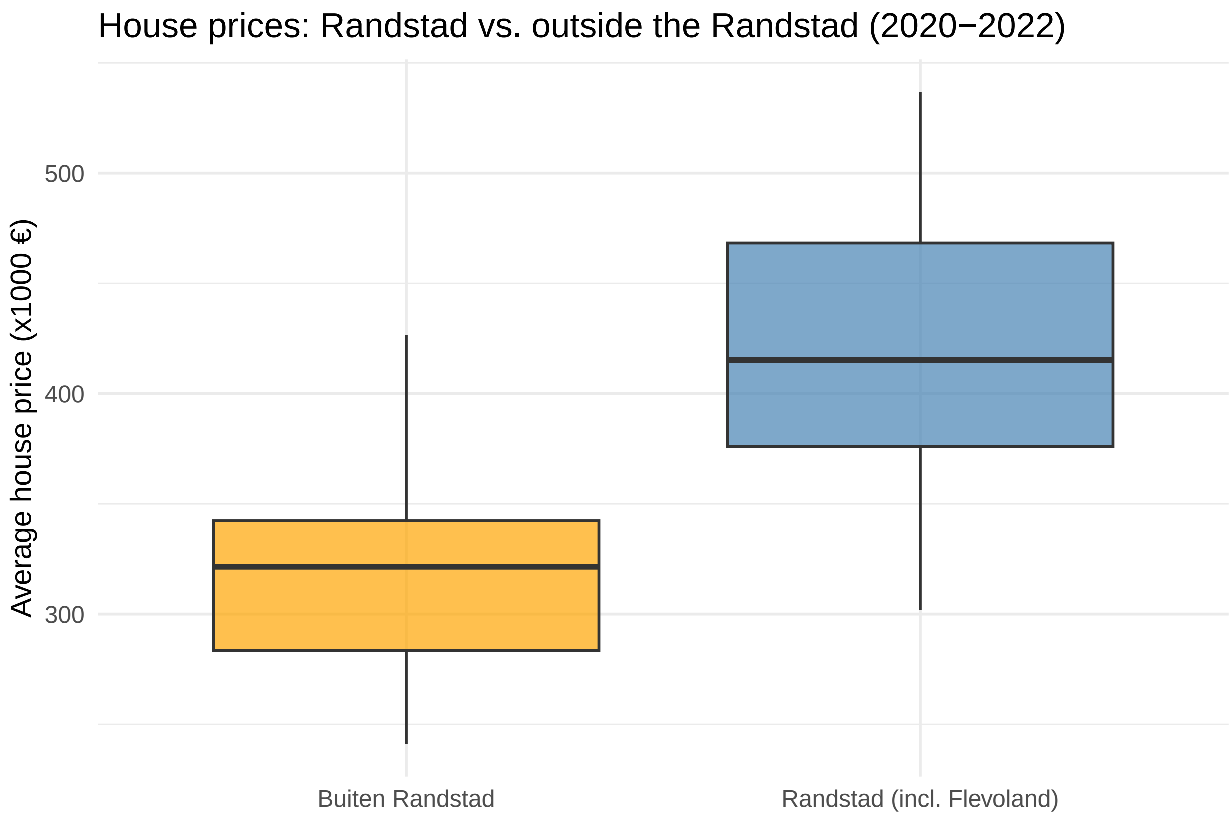
For the spatial variation we loaded the province boundaries and merged the average income and the average house price per province onto the map, which gave us two maps. We placed these next to each other as a single figure with panels (a) and (b), using the `patchwork` package, so that the two patterns can be compared directly.



The two maps show the spatial distribution of (a) average disposable income and (b) the average house price per province. The highest incomes and the highest house prices both concentrate in the same central provinces (Utrecht and Noord-Holland), while the peripheral provinces (Groningen, Zeeland, Drenthe) score low on both. However, house prices rise more steeply than income across the map, so the regions that are already expensive become relatively even less accessible. This spatial inequality in access to housing is the core of our social problem.

3.5 Visualize sub-population variation

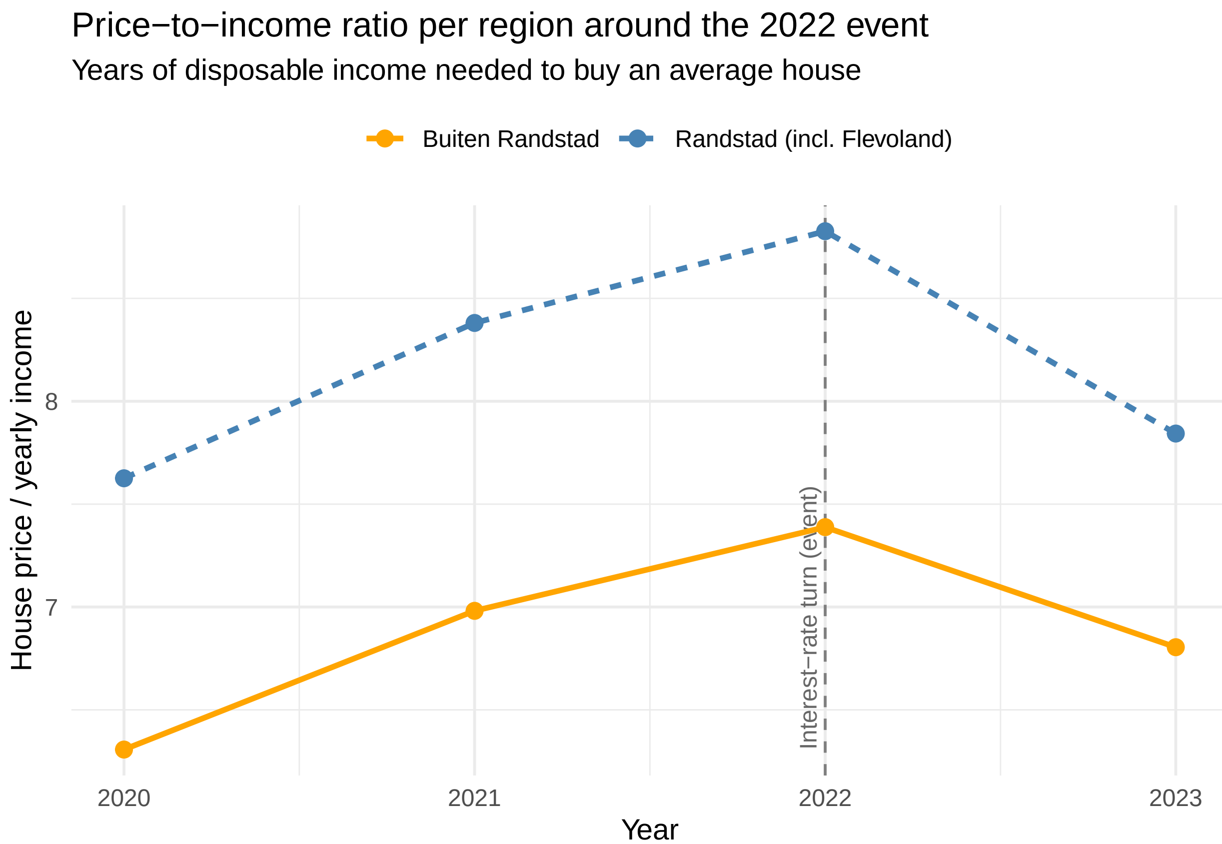
To look at the sub-population variation we took the house price per province and year and labelled each province as either Randstad (Noord-Holland, Zuid-Holland, Utrecht and Flevoland) or outside the Randstad, and then compared the two groups with a boxplot. A boxplot shows the whole distribution of house prices within each group (median, spread and outliers) instead of twelve separate provinces.



This boxplot compares the distribution of average house prices in the Randstad with the rest of the country. Prices in the Randstad are clearly higher: both the median and the whole range sit above those outside the Randstad. This is relevant to our social problem because it shows that the affordability pressure is concentrated in exactly the densely populated provinces where most people want or need to live, where housing is least affordable.

3.6 Event analysis

For the event analysis we study the relationship between two variables – average disposable household income and the average house price around a clear event: the turn in mortgage interest rates in 2022. To do this we generate a fourth variable, the **price-to-income ratio** (average house price divided by average yearly disposable income), which can be read as "the number of years of income needed to buy an average house". The plot below shows this ratio over time for the Randstad and the rest of the country, with the 2022 event marked.



This event analysis is central to our social problem. Between 2020 and 2022 the price-to-income ratio rose in both regions, from about 7.6 to 8.8 years of income in the Randstad and from 6.3 to 7.4 outside it, meaning houses became steadily less affordable relative to local income. After the 2022 interest-rate turn the ratio falls again in 2023 (prices declined by roughly 1% to 5% while incomes kept rising by about 7%). The fact that affordability moves with the financing event rather than with local income growth suggests that house prices in this period were driven more by mortgage conditions than by regional purchasing power – and that the affordability gap is structurally largest in the Randstad.

Part 4 - Discussion

4.1 Discuss your findings

Our analysis points to one consistent pattern: across Dutch provinces, house prices have grown considerably faster than disposable income, so housing has become less affordable almost everywhere. The decoupling is strongest in the Randstad, where the price-to-income ratio is both the highest and the fastest-rising, but the peripheral provinces follow the same direction. The spatial analysis shows that high prices cluster in the same densely populated, high-income provinces, yet because prices outrun income, even relatively wealthy regions face an affordability gap.

The event analysis adds an important nuance. The affordability ratio peaks in 2022 and then falls in 2023, mirroring the turn in mortgage interest rates rather than any change in local income. This suggests that the recent price dynamics were driven more by financing conditions than by regional purchasing power, which means affordability could deteriorate again quickly if borrowing becomes cheaper.

Several limitations should be kept in mind. The overlapping window in which we have both income and house-price data per province is short (2020–2023), so the trends rest on few observations and should be read as indicative rather than conclusive. Administrative income definitions can change between years, which may affect comparability, and the analysis covers only the twelve provinces, so within-province differences (e.g. between cities and rural areas) are hidden. Finally, our results show association, not causation: income and house prices move together, but both are likely influenced by common drivers such as interest rates and the supply of housing.

Part 5 - Reproducibility

5.1 Github repository link

Provide the link to your PUBLIC repository here: https://github.com/Kian-Hendriks/R-project-group-3-3

5.2 Reference list

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